REV A LAYOUT REVIEW. NOT FOR FABRICATION.

Four–slot carrier: protected external 5 V / 4 A, isolated GPIO

Local labels connect equal names. NC marks deliberate unused module pins.
Selected parts: see BOM, FOOTPRINT_REVIEW.md and ASSEMBLY_NOTES.md.

R20/R21: immediately after U5 SCK/MOSI outputs. R22: at U4 SO before shared MISO.
22R initial; 0R population option. Verify by oscilloscope/IBIS. No long stubs.
Each radio card has its own MISO source resistor before slot A7.

18/26 GPIO used; eight spare including GP26–28, all physically NC.
BUSY/IRQ/CS are direct GPIO through fixed–direction isolation, not expander GPIO.
KEY: captive M3 x 25 alignment post, see radio–slot–v0.1.md.
U4 GPA0..5 = S1/S3/S4 TX,RX pairs; GPB0..3 = S1..4 RESET_N.
No autonomous duty–cycled RX or automatic TX/RX transitions on Ebyte cards.

USB–only: Q1 off; RADIO_5V and IOREF_3V3 unpowered. External power: U2 –> radios and Q1 –> VSYS.
Q1 orientation is intentional: body diode from RADIO_5V toward VSYS. Pico onboard D1 blocks USB backfeed.
#FLG01/#FLG02 model the external supply; #FLG03 models real Q1/onboard USB–D1 ORing output.
U2 AUXOFF/FLT/ITIMER unused per datasheet. Leave ITIMER open for fastest current limiting.

SCHEMATIC REVIEW – NOT FOR FABRICATION

Sheet: /
File: carrier–rev–a.kicad_sch

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